



PES UNIVERSITY, BANGALORE

Department of Computer Science and Engineering



Software Requirements Specification

Vehicle Parking System

Version 1.0

Prepared by Team 10, Section C

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Revision History

Name	Date	Reason For Changes	Version
Team 10, Section C	05-10-2026	First complete version for the Part-1 deliverable	1.0

Project Team

Name	SRN	Role	Responsibility
Vivian Sobers E	PES1UG24CS901	Integration and DevOps	Repository, CI pipeline, deployment, integration of the web application with the forecast service
D Harikrishnan	PES1UG24CS136	Database	Schema design, migrations, constraints, indexing, data integrity and reporting queries
Dhruv Purushotham Raju	PES1UG24CS154	Frontend	Templates, HTMX interactions, forms, accessibility and responsive layout
Gitanjali A	PES1UG24CS170	Backend	Django services, REST endpoints, business rules, allocation and billing logic

1. Introduction

1.1 Purpose

This document specifies the software requirements for release 1.0 of the Vehicle Parking System (VPS), a web application for booking parking slots, running the entry and exit gates of a car park, billing stays and reporting on use. It covers the whole system: the web application used by drivers, gate attendants and lot administrators, and the separate forecast service that predicts slot availability. It is the reference for the Software Test Plan and the Software Architecture and Design Specification of the same release.

1.2 Intended Audience and Reading Suggestions

The readers are the four developers of Team 10, the course instructor who evaluates the project, and anyone who later maintains or tests the system.

- Developers should read Sections 2, 4 and 5 for what to build and Section 6 for the limits it must respect.
- Testers should read Sections 5 and 6; every requirement has an identifier that the test plan traces to.
- Evaluators can start with Section 1.3 for the scope, Section 4 for the use case model and Appendix C for traceability.

Section 2 describes the product in general terms. Section 3 covers external interfaces, Section 4 the analysis models, Section 5 the functional requirements grouped by feature, Section 6 the non-functional requirements including security, and Section 7 the remaining requirements. The appendices hold the glossary, field layouts and the requirement traceability matrix.

1.3 Product Scope

Many car parks still work on first come, first served: drivers circle looking for a bay, attendants keep paper logs, and nobody knows how full the car park will be in an hour. VPS lets a driver reserve a slot that suits their vehicle for a chosen time window, guarantees that no slot is promised to two drivers at once, and gives attendants a quick way to check vehicles in and out with correct charges.

The objectives of this release are:

- No double bookings, enforced by a database constraint in addition to application checks.
- A receipt for every stay whose charge can be recomputed exactly from stored data.
- Gate actions completed in under 2 seconds so that queues do not form at the barrier.
- Occupancy and revenue reports for lot administrators.
- An advisory forecast of whether a slot will be free at a future arrival time.

Out of scope for this release: online payment, physical barrier and camera hardware, number-plate recognition, mobile apps, multi-operator marketplaces and GST invoicing.

1.4 Definitions, Acronyms and Abbreviations

Terms specific to this document are defined in Appendix A. The requirement identifier prefixes are: FR (functional requirement), NFR (non-functional requirement), SAF (safety), SO (security objective), SEC (security requirement), BR (business rule), DR (domain requirement) and OR (other requirement).

1.5 References

- IEEE Std 830-1998, IEEE Recommended Practice for Software Requirements Specifications.
- PES University, Software Engineering (UE24CS341A) Mini-Project Deliverables Part-1, 2026.
- VPS Software Test Plan, version 1.0, 05-10-2026 (docs/part-1 in the project repository).
- VPS Software Architecture and Design Specification, version 1.0, 05-10-2026 (same location).
- OWASP Application Security Verification Standard 4.0.3, <https://owasp.org/www-project-application-security-verification-standard/>
- Web Content Accessibility Guidelines (WCAG) 2.1, W3C Recommendation, <https://www.w3.org/TR/WCAG21/>
- Digital Personal Data Protection Act, 2023, Government of India.
- Central Motor Vehicles Rules, 1989 (vehicle registration marks), Government of India.
- Project repository: <https://github.com/VivianSobers/Vehicle-Parking-System>

2. Overall Description

2.1 Product Perspective

VPS is a new, self-contained product. It does not replace an earlier system, although it replaces the paper logs and walk-up allocation that many car parks use today. It consists of a Django web application backed by PostgreSQL, a scheduler for time-based jobs, and a separate FastAPI service that serves availability forecasts. All users work through a web browser. The only external systems are an SMTP mail server for e-mail and, in deployment, an nginx reverse proxy that terminates TLS. The component structure is described in the Software Architecture and Design Specification.

2.2 Product Functions

The major functions, detailed in Section 5, are:

- **User Accounts and Access Control (AUTH):** Drivers create their own accounts.
- **Vehicle Registry (VEH):** Drivers keep a list of their vehicles so that bookings and gate checks refer to a known registration number and vehicle type.
- **Lot and Slot Management (LOT):** Lot Administrators describe each car park: its opening hours, its slots, the type of each slot and how far each slot is from the exit.
- **Slot Availability Search (SRCH):** Drivers check whether a slot suitable for their vehicle is free for the whole period they need, and anyone can see the live occupancy of a lot.
- **Reservation Management (RES):** A reservation holds one slot for one vehicle over one time window.
- **Slot Allocation (ALLOC):** The system, not the Driver, chooses the slot.
- **Gate Check-in and Check-out (GATE):** Gate Attendants record vehicles entering and leaving, including vehicles that arrive without a booking.
- **Pricing and Billing (BILL):** Lot Administrators set hourly tariffs.
- **Reports and Audit (RPT):** Lot Administrators see how the car parks are used and what they earn, and can review privileged actions.
- **Occupancy Forecast (FCST):** Drivers planning ahead see how likely it is that a slot will be free when they arrive.

2.3 User Classes and Characteristics

User class	Characteristics	Main functions	Importance
Driver	Members of the public; any level of technical skill; mostly on mobile phones; use the system a few times a week.	Register, manage vehicles, search, reserve, cancel, view receipts and forecasts.	Primary
Gate Attendant	Car park staff working at the entry and exit; trained for 1 hour; use a desktop or tablet all shift and need speed.	Check in, check out, register walk-ins, record payments, correct times after an outage.	Primary

User class	Characteristics	Main functions	Importance
Lot Administrator	Car park manager; comfortable with forms and spreadsheets; uses the system daily.	Configure lots, slots and tariffs; manage staff; view reports and the audit log.	Secondary
System Clock	Time-based actor, not a person.	Expires unclaimed reservations; triggers weekly forecast retraining.	Supporting

2.4 Operating Environment

- Server: Linux (Ubuntu 22.04 LTS or later) with Python 3.12, PostgreSQL 16 with the btree_gist extension, and nginx.
- Application: Django 5.x with Django REST Framework, served by Gunicorn; forecast service on FastAPI with scikit-learn.
- Clients: the two latest versions of Chrome, Firefox, Edge and Safari, on desktop and on phones 360 pixels wide or more.
- Development and CI: GitHub, GitHub Actions and GitHub Projects.

2.5 Design and Implementation Constraints

- Python is the only implementation language for server code; the user interface uses server-rendered Django templates with HTMX, with no separate JavaScript framework.
- PostgreSQL is required because the no-double-booking rule relies on an exclusion constraint over time ranges.
- The course requires an agile process: work is planned in GitHub Projects, every change goes through a pull request with review, and CI must pass before merge.
- Automated tests use pytest; coverage is measured in CI (NFR-MNT-01).
- Each team member owns one area: Vivian Sobers E (integration and DevOps), D Harikrishnan (database), Dhruv Purushotham Raju (frontend) and Gitanjali A (backend).
- No payment data is processed; this keeps the system outside PCI DSS scope.

2.6 Assumptions and Dependencies

- Each slot has a fixed type and a fixed distance rank set by the administrator; the system has no sensors and relies on attendants for entry and exit times.
- Drivers have an e-mail address and a mobile phone; confirmations are sent by e-mail only.
- An SMTP account is available for sending e-mail. If it is not, bookings still work and e-mails are retried.
- There is no historical booking data at launch, so the forecast uses a baseline from simulated arrivals until 4 weeks of real data exist (FR-FCST-02).
- Fees and time limits in this document (for example the Rs 30 late-cancellation charge and the 15-minute hold) are defaults chosen by the team and can be changed by configuration.

3. External Interface Requirements

3.1 User Interfaces

All interfaces are web pages built with Django templates. Common rules:

- Every page has a top bar with the product name, the signed-in user's name and role, and a Sign out link; a Help link opens a short page describing the current screen.
- Forms validate on the server and show errors next to the field in red text with an icon, in plain language that states how to fix the problem.
- Success messages appear in a green banner at the top of the page and disappear after 5 seconds.
- Times are entered and shown in 24-hour IST; dates as DD-MM-YYYY.
- Layouts adapt from 360-pixel phones to desktop screens; tap targets are at least 44 by 44 pixels.
- The gate console accepts keyboard-only use: the identifier field has focus on load and Enter submits.

Main screens: Driver home (unpaid total, upcoming reservations), Vehicles, Search and book (with forecast panel), Booking confirmation, Booking history, Receipt; Gate console (check-in, check-out, walk-in, payment); Administration (lots, slots, blocks, tariffs, staff); Reports; Audit log.

3.2 Software Interfaces

Interface	Version	Data exchanged and purpose
PostgreSQL	16	All persistent data through the Django ORM; the btree_gist extension provides the exclusion constraint for FR-RES-02.
Forecast service (internal)	VPS 1.0	REST/JSON over HTTP on the private network. Request: lot id, vehicle type, arrival time. Response: probability and band. Timeout 2 seconds (FR-FCST-03).
SMTP mail server	Any, with STARTTLS	Outgoing booking, cancellation, move, expiry and password-reset e-mails. No incoming mail.
Django and Django REST Framework	5.x and 3.15	Web framework, authentication, sessions, CSRF protection, admin and JSON endpoints.
HTMX	2.x	Partial page updates for search results and the live occupancy view.

3.3 Communications Interfaces

- Browsers talk to the server over HTTPS (TLS 1.2 or later) on port 443; port 80 only redirects to HTTPS (SEC-04).
- The web application calls the forecast service over HTTP on a private network that is not reachable from the internet.

- JSON is encoded as UTF-8; times use ISO 8601 with an offset, for example 2026-10-07T09:30:00+05:30.
- E-mail uses SMTP with STARTTLS on port 587.

3.4 Hardware Interfaces

This release has no direct hardware interfaces. Gate barriers and cameras are operated separately; attendants use any device with a supported browser.

4. Analysis Models

4.1 Use Case Diagram

Figure 1 shows the actors and use cases of VPS. Driver, Gate Attendant and Lot Administrator are specialisations of User, so all three can log in and out. Reserve slot and Register walk-in vehicle both include Allocate slot; Check out vehicle and Cancel reservation include Compute parking charge; View predicted availability extends Search slot availability.



Figure 1: Use case diagram of the Vehicle Parking System

4.2 Use Case Descriptions

Use case	Actor	Main flow	Alternative flows	Requirements
Reserve slot	Driver	1. Choose lot, vehicle and window. 2. System shows eligible free slots. 3. Driver confirms. 4. System allocates a	Window invalid: message names the rule. No slot free, limit reached or unpaid	FR-SRCH-01, FR-RES-01 to FR-RES-03, FR-

Use case	Actor	Main flow	Alternative flows	Requirements
		slot and shows booking reference and slot code.	charge: booking refused with reason.	ALLOC-01 to 03
Check in vehicle	Gate Attendant	1. Enter booking reference or registration number. 2. System checks state and window. 3. System marks reservation Active and shows slot.	Too early: refused with earliest time. Slot still occupied: new slot allocated.	FR-GATE-01, FR-GATE-02, FR-GATE-07
Check out vehicle	Gate Attendant	1. Enter registration number. 2. System computes charge and shows receipt. 3. Attendant records payment.	No active stay: error. Payment later: charge stays unpaid on the Driver's account.	FR-GATE-04, FR-GATE-05, FR-BILL-02 to 04
Expire unclaimed reservations	System Clock	1. Every minute, find Reserved bookings more than 15 minutes past start. 2. Mark Expired, free slot, add no-show charge, e-mail Driver.	Mail server down: e-mail retried later.	FR-RES-05, FR-RES-07

4.3 Reservation State Model

Figure 2 shows the states of a reservation. Only Reserved and Active reservations hold a slot; the exclusion constraint behind FR-RES-02 applies to these two states only.

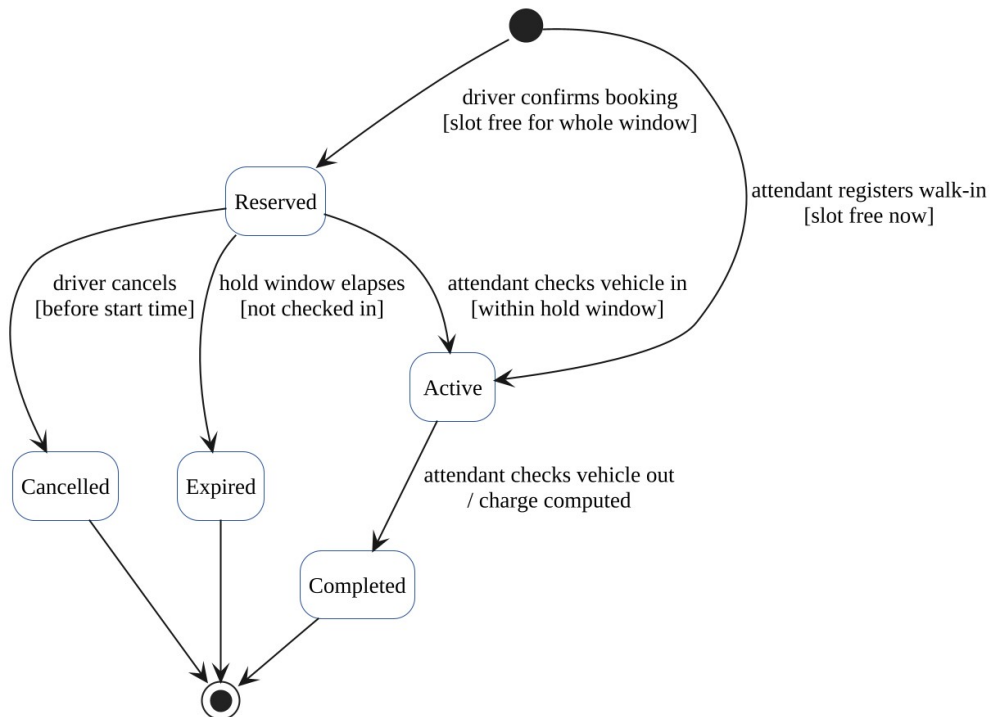


Figure 2: State diagram of a reservation

5. System Features

Functional requirements are grouped by feature. Each requirement has a unique identifier used by the test plan and the traceability matrix in Appendix C. Priority ratings use a 1 (low) to 9 (high) scale.

5.1 User Accounts and Access Control

5.1.1 Description and Priority

Drivers create their own accounts. Gate attendants and lot administrators are created by an administrator. Every user signs in with an e-mail address and password and sees only the functions allowed for their role. Priority: **High** (Benefit 9, Penalty 9, Cost 4, Risk 3).

5.1.2 Stimulus/Response Sequences

Stimulus	System response
A visitor submits the registration form.	The system validates the fields, creates a Driver account and signs the user in.
A user submits an e-mail address and password.	The system signs the user in and opens the home page for the user's role, or shows a generic error.
A user requests a password reset.	The system e-mails a one-time reset link.

5.1.3 Functional Requirements

ID	Requirement
FR-AUTH-01	The system shall let a visitor register as a Driver by entering full name (2 to 100 characters), e-mail address, a 10-digit Indian mobile number starting with 6, 7, 8 or 9, and a password. The e-mail address shall be unique; a duplicate shall be rejected with the message "An account with this e-mail already exists".
FR-AUTH-02	The system shall authenticate users by e-mail address and password and shall let any signed-in user log out.
FR-AUTH-03	The system shall support exactly three roles: Driver, Gate Attendant and Lot Administrator. Gate Attendant and Lot Administrator accounts shall be created only by a Lot Administrator.
FR-AUTH-04	The system shall send a password reset link to a registered e-mail address. The link shall be single-use and expire 30 minutes after it is issued.
FR-AUTH-05	A Lot Administrator shall be able to deactivate and reactivate any staff account. A deactivated account shall not be able to sign in, and its active sessions shall end within 1 minute.
FR-AUTH-06	A Lot Administrator shall be able to mark a Driver as an accessible-parking permit holder after checking the permit offline.

5.2 Vehicle Registry

5.2.1 Description and Priority

Drivers keep a list of their vehicles so that bookings and gate checks refer to a known registration number and vehicle type. Priority: **High** (Benefit 7, Penalty 6, Cost 2, Risk 2).

5.2.2 Stimulus/Response Sequences

Stimulus	System response
A Driver adds a vehicle.	The system validates the registration number and type and adds the vehicle to the Driver's list.
A Driver removes a vehicle.	The system removes it, or refuses if the vehicle has an upcoming or active reservation.

5.2.3 Functional Requirements

ID	Requirement
FR-VEH-01	The system shall let a Driver register up to 5 vehicles, each with a registration number, a vehicle type (Car, Two-wheeler or Electric car) and an optional nickname of up to 30 characters.
FR-VEH-02	The system shall store registration numbers in upper case without spaces and shall accept only the standard Indian format (for example KA01AB1234) or the Bharat series format (for example 22BH1234AA).
FR-VEH-03	A registration number shall be unique among active vehicles in the system.
FR-VEH-04	The system shall refuse to remove a vehicle that has a reservation in the Reserved or Active state.

5.3 Lot and Slot Management

5.3.1 Description and Priority

Lot Administrators describe each car park: its opening hours, its slots, the type of each slot and how far each slot is from the exit. Priority: **High** (Benefit 8, Penalty 8, Cost 3, Risk 2).

5.3.2 Stimulus/Response Sequences

Stimulus	System response
An administrator creates a slot.	The system records the slot with its type and distance rank and includes it in searches.
An administrator blocks a slot for maintenance.	The system removes the slot from search and allocation for the block period and moves affected reservations.

5.3.3 Functional Requirements

ID	Requirement
FR-LOT-01	A Lot Administrator shall be able to create, edit and archive parking lots with a name, address, opening time and closing time.
FR-LOT-02	A Lot Administrator shall be able to create slots in a lot. Each slot shall have a code unique within the lot, a slot type (Standard, Two-wheeler, EV charging or Accessible) and a distance rank (a positive integer; 1 is nearest to the exit).
FR-LOT-03	A Lot Administrator shall be able to block a slot for a stated period with a reason. A blocked slot shall not appear in search results or be allocated during that period.
FR-LOT-04	When a block overlaps existing Reserved reservations, the system shall move each one to another eligible slot free for the same window and e-mail the Driver the new slot code. If no slot is free, the system shall cancel the reservation without any fee and notify the Driver.

5.4 Slot Availability Search

5.4.1 Description and Priority

Drivers check whether a slot suitable for their vehicle is free for the whole period they need, and anyone can see the live occupancy of a lot. Priority: **High** (Benefit 9, Penalty 7, Cost 3, Risk 3).

5.4.2 Stimulus/Response Sequences

Stimulus	System response
A Driver enters a lot, a vehicle and a time window.	The system shows the number of eligible free slots for the whole window, or a validation message.
A user opens a lot's occupancy page.	The system shows free, reserved, occupied and blocked counts and refreshes them.

5.4.3 Functional Requirements

ID	Requirement
FR-SRCH-01	Given a lot, one of the Driver's vehicles, a start time and an end time, the system shall show the number of slots that are eligible for that vehicle and free for the entire window.
FR-SRCH-02	The system shall reject a search window that starts in the past, starts more than 7 days ahead, is shorter than 30 minutes, is longer than 24 hours or falls outside the lot's opening hours, and shall state which rule was broken.
FR-SRCH-03	The system shall show, for each lot, the count of free, reserved, occupied and blocked slots by slot type, refreshed automatically at least every 30 seconds while the page is open.

5.5 Reservation Management

5.5.1 Description and Priority

A reservation holds one slot for one vehicle over one time window. It moves through the states Reserved, Active, Completed, Cancelled and Expired (Figure 2). Two reservations whose windows overlap can never hold the same slot. Priority: **High** (Benefit 9, Penalty 9, Cost 6, Risk 7).

5.5.2 Stimulus/Response Sequences

Stimulus	System response
A Driver confirms a booking.	The system allocates a slot, creates the reservation and shows the booking reference and slot code.
A Driver cancels a reservation.	The system cancels it, frees the slot and records a late-cancellation fee if the rule applies.
The hold window passes without a check-in.	The system expires the reservation, frees the slot and records a no-show fee.

5.5.3 Functional Requirements

ID	Requirement
FR-RES-01	When a Driver confirms a valid search, the system shall allocate a slot (FR-ALLOC-02), create a reservation in the Reserved state and show a booking reference in the form VPS-YYMMDD-NNNN together with the slot code.
FR-RES-02	The system shall never let two reservations in the Reserved or Active state hold the same slot for overlapping windows. When several requests compete for the last eligible slot, exactly one shall succeed and the others shall receive the message "No slot available for this window".
FR-RES-03	The system shall refuse a new reservation if the Driver already has 2 reservations in the Reserved state or has any unpaid charge, and shall state the reason.
FR-RES-04	A Driver shall be able to cancel a Reserved reservation before its start time. Cancelling 60 minutes or more before the start shall be free; cancelling later shall record a late-cancellation charge of Rs 30.
FR-RES-05	If a Reserved reservation is not checked in within 15 minutes after its start time, the system shall change it to Expired, free the slot and record a no-show charge of Rs 50. The expiry job shall run at least once every minute.
FR-RES-06	The system shall show each Driver their reservations, newest first, 20 per page, with booking reference, lot, slot, window, state and charges.
FR-RES-07	The system shall e-mail the Driver within 2 minutes of a reservation being created, cancelled, moved or expired.

5.6 Slot Allocation

5.6.1 Description and Priority

The system, not the Driver, chooses the slot. The rule is fixed and deterministic so that the same request on the same data always gives the same slot. Priority: **High** (Benefit 8, Penalty 7, Cost 5, Risk 5).

5.6.2 Stimulus/Response Sequences

Stimulus	System response
A reservation or walk-in needs a slot.	The system returns the best eligible slot that is free for the window, or reports that none is free.

5.6.3 Functional Requirements

ID	Requirement
FR-ALLOC-01	The system shall treat a slot as eligible for a vehicle as follows: Two-wheeler vehicles use Two-wheeler slots only; Cars use Standard slots, and Accessible slots only when the Driver holds an accessible-parking permit; Electric cars use EV charging slots first and Standard slots when no EV charging slot is free.
FR-ALLOC-02	Among eligible slots free for the whole window and not blocked, the system shall choose the slot with the preferred type first, then the lowest distance rank, then the lowest slot code.
FR-ALLOC-03	If the chosen slot is taken by a concurrent request, the system shall try the next candidate, up to 3 attempts in total, before reporting that no slot is available.

5.7 Gate Check-in and Check-out

5.7.1 Description and Priority

Gate Attendants record vehicles entering and leaving, including vehicles that arrive without a booking. Priority: **High** (Benefit 9, Penalty 9, Cost 5, Risk 5).

5.7.2 Stimulus/Response Sequences

Stimulus	System response
An attendant enters a booking reference or registration number at entry.	The system checks the booking and the time window, then marks the reservation Active.
A vehicle without a booking arrives.	The attendant registers a walk-in and the system allocates a slot.
An attendant enters a registration number at exit.	The system computes the charge, shows the receipt and frees the slot.

5.7.3 Functional Requirements

ID	Requirement
FR-GATE-01	Given a booking reference or registration number, the system shall check in a Reserved reservation only between 15 minutes before and 15 minutes after its start time, record the entry time and change the state to Active.
FR-GATE-02	For an early arrival, the system shall refuse check-in and show the earliest time at which check-in is allowed.
FR-GATE-03	A Gate Attendant shall be able to register a walk-in vehicle by registration number and vehicle type. The system shall allocate a slot that is free from now until the lot's closing time and create an Active walk-in reservation, or report that no such slot exists.
FR-GATE-04	Given a registration number or booking reference of an Active reservation, the system shall record the exit time, compute the charge (FR-BILL-02, FR-BILL-03), show an itemised receipt, change the state to Completed and free the slot immediately.
FR-GATE-05	A Gate Attendant shall be able to record a payment against a charge with the method Cash, UPI or Card.
FR-GATE-06	When the system was unavailable, a Gate Attendant shall be able to enter an entry or exit time up to 24 hours in the past with a mandatory reason; each such correction shall be written to the audit log.
FR-GATE-07	If, at check-in, the allocated slot is still occupied by an overstaying vehicle, the system shall allocate another eligible free slot for the same window and show the new slot code.

5.8 Pricing and Billing

5.8.1 Description and Priority

Lot Administrators set hourly tariffs. The system calculates every charge from stored inputs so that any receipt can be recomputed exactly. Priority: **High** (Benefit 8, Penalty 8, Cost 4, Risk 4).

5.8.2 Stimulus/Response Sequences

Stimulus	System response
An administrator saves a tariff.	The system records it with its effective date.
A vehicle checks out.	The system computes the parking and overstay charge and produces a receipt.

5.8.3 Functional Requirements

ID	Requirement
FR-BILL-01	A Lot Administrator shall be able to set an hourly rate in rupees for each lot and slot type, with an effective-from date. A charge shall use the tariff in force at the vehicle's entry time.

ID	Requirement
FR-BILL-02	The parking charge shall be the hourly rate multiplied by the billable hours. The billable period runs from entry to the reserved end time or the exit time, whichever is earlier (to the exit time for walk-ins); it is rounded up to the next whole 30 minutes, with a minimum of 1 hour.
FR-BILL-03	If a vehicle leaves more than 10 minutes after the reserved end time, the time from the end of that 10-minute grace period to the exit shall be billed as overstay at 1.5 times the hourly rate, using the same 30-minute rounding. Walk-in stays have no overstay.
FR-BILL-04	Each receipt shall show the booking reference, registration number, slot, entry and exit times, billable duration, rate, each charge line, total, payment method and payment status, and shall be printable from the browser.
FR-BILL-05	The system shall show each Driver the total of their unpaid charges on their home page.

5.9 Reports and Audit

5.9.1 Description and Priority

Lot Administrators see how the car parks are used and what they earn, and can review privileged actions. Priority: **Medium** (Benefit 6, Penalty 4, Cost 4, Risk 2).

5.9.2 Stimulus/Response Sequences

Stimulus	System response
An administrator chooses a report and a date range.	The system shows the report and offers a CSV download.

5.9.3 Functional Requirements

ID	Requirement
FR-RPT-01	The system shall show Lot Administrators the current occupancy of each lot by slot type.
FR-RPT-02	The system shall produce an occupancy report for a chosen lot and date range of up to 31 days, giving the average occupancy percentage for each hour, and shall export it as CSV.
FR-RPT-03	The system shall produce a revenue report for a chosen date range of up to 31 days, with totals by lot, slot type and payment method, and shall export it as CSV.
FR-RPT-04	The system shall let Lot Administrators search the audit log by actor, action and date range.

5.10 Occupancy Forecast

5.10.1 Description and Priority

Drivers planning ahead see how likely it is that a slot will be free when they arrive. The forecast is advisory; the booking rules do not depend on it. Priority: **Low** (Benefit 5, Penalty 2, Cost 6, Risk 6).

5.10.2 Stimulus/Response Sequences

Stimulus	System response
A Driver views availability for a future arrival time.	The system shows the predicted chance that an eligible slot is free, or a notice that no forecast is available.

5.10.3 Functional Requirements

ID	Requirement
FR-FCST-01	For a lot, a vehicle type and an arrival time up to 24 hours ahead, the system shall show the predicted probability that at least one eligible slot is free, as a percentage and as a band (Low below 40%, Medium 40% to 75%, High above 75%).
FR-FCST-02	The forecast model shall be retrained once a week from completed reservations. Until 4 weeks of history exist, the system shall use a baseline built from simulated arrivals and label the result "Estimate".
FR-FCST-03	If the forecast service does not answer within 2 seconds, availability search shall still work and the forecast panel shall show "Forecast not available".

6. Other Nonfunctional Requirements

6.1 Performance Requirements

Targets are measured on the staging server (2 vCPU, 4 GB RAM) with the test data described in the test plan.

ID	Requirement
NFR-PERF-01	Availability search (FR-SRCH-01) shall respond within 1 second at the 95th percentile for a lot of 500 slots with 50 concurrent users.
NFR-PERF-02	Creating a reservation (FR-RES-01) and check-in or check-out (FR-GATE-01, FR-GATE-04) shall each complete within 2 seconds at the 95th percentile under the same load.
NFR-PERF-03	The system shall support 200 concurrent signed-in sessions without errors caused by load.
NFR-PERF-04	Each page shall finish loading within 3 seconds on a 10 Mbps connection, with no more than 500 KB transferred on first load.
NFR-PERF-05	The forecast service shall answer within 500 milliseconds at the 95th percentile.

6.2 Safety Requirements

VPS does not control machinery, but wrong information can send drivers to closed bays or leave vehicles stuck at the gate.

ID	Requirement
SAF-01	The system shall never allocate or show as free a slot that is blocked for maintenance, so that vehicles are not sent to unsafe or closed bays.
SAF-02	Accessible slots shall be allocated only to Drivers holding an accessible-parking permit, so that they remain free for people who need them.
SAF-03	If the application is unavailable, gate operations shall continue on a paper log, and attendants shall be able to enter the recorded times later (FR-GATE-06), so that no vehicle is held at the gate because of a system fault.

6.3 Security Requirements

6.3.1 Security Objectives

ID	Objective	Statement
SO-01	Confidentiality of personal data	Driver names, e-mail addresses, mobile numbers and vehicle registration numbers are seen only by the Driver and by staff who need them for their work.
SO-02	Integrity of reservations and charges	No user can create, change or delete reservations, tariffs or charges beyond what their role allows, and every charge can be recomputed from stored data.
SO-03	Accountability for	Every action by a Gate Attendant or Lot Administrator that

ID	Objective	Statement
	privileged actions	changes money, slots or accounts can be traced to the person who performed it.
SO-04	Availability under misuse	Password guessing and request flooding from a single client cannot lock out other users or make the service unusable.

6.3.2 Security Requirements

Each requirement names the objectives it supports. The test plan (Section 5.1) describes how each is validated.

ID	Objectives	Requirement
SEC-01	SO-01, SO-02	Passwords shall be stored only as salted PBKDF2-SHA256 hashes using Django's default iteration count, and shall be at least 8 characters long and contain at least one letter and one digit.
SEC-02	SO-04	After 5 failed sign-in attempts for one account within 15 minutes, the account shall be locked for 15 minutes, and sign-in attempts shall be limited to 10 per minute per IP address.
SEC-03	SO-02	Every request shall be checked on the server against the role permission matrix (Appendix B.3). A request for a function outside the user's role shall receive HTTP 403 and be logged.
SEC-04	SO-01	In deployment all traffic shall use HTTPS with TLS 1.2 or later; plain HTTP shall redirect to HTTPS, and session cookies shall be Secure, HttpOnly and SameSite=Lax.
SEC-05	SO-02	Every state-changing request shall require a valid CSRF token, and a request without one shall be rejected with HTTP 403.
SEC-06	SO-01, SO-02	All input shall be validated on the server, and database access shall use the ORM's parameterised queries only; SQL shall never be built by joining strings.
SEC-07	SO-03	The audit log shall record actor, action, target, old value, new value and timestamp for every tariff change, slot block, staff account change, payment, charge waiver and manual time correction. Application users shall not be able to edit or delete audit entries, which shall be kept for at least 1 year.
SEC-08	SO-01	A signed-in session shall end after 30 minutes without activity, and the session identifier shall be rotated at sign-in.
SEC-09	SO-01	Gate Attendants shall see a Driver's mobile number masked except for the last 4 digits, and no screen or log shall show a password, password hash or reset token.
SEC-10	SO-02	Secrets such as the Django secret key, database password and SMTP password shall be read from environment variables and shall never be committed to the repository.

6.4 Software Quality Attributes

ID	Attribute	Requirement
NFR-REL-01	Reliability	Under 50 simultaneous booking requests for the last free slot, the system shall create exactly one reservation in every test run (see FR-RES-02).
NFR-AVL-01	Availability	The deployed system shall be available at least 99% of the time between 06:00 and 23:00 IST, measured per calendar month.
NFR-USE-01	Usability	A first-time Driver shall be able to register a vehicle and complete a reservation in under 3 minutes without help, and booking shall take no more than 4 screens.
NFR-USE-02	Usability	All pages shall work on screens 360 pixels wide and larger and shall meet WCAG 2.1 level AA colour contrast.
NFR-MNT-01	Maintainability	Backend code shall have at least 70% line coverage and 60% branch coverage from automated tests, measured in CI on every push.
NFR-MNT-02	Maintainability	Code shall pass the ruff linter with no errors, and a change shall reach the main branch only through a reviewed pull request with a passing CI run.
NFR-PRT-01	Portability	The system shall run on any Linux server with Python 3.12 and PostgreSQL 16, and shall work in the two latest versions of Chrome, Firefox, Edge and Safari.
NFR-TST-01	Testability	Business rules for allocation and billing shall be callable without a web request so that they can be unit-tested directly.

Where attributes conflict, the team gives priority in this order: correctness of bookings and charges, security, gate speed, then ease of use for drivers.

6.5 Business Rules

ID	Rule
BR-01	Only Lot Administrators can change tariffs, slots, lots and staff accounts.
BR-02	Only Gate Attendants check vehicles in and out, register walk-ins and record payments.
BR-03	A Driver may hold at most 2 reservations in the Reserved state at a time.
BR-04	A Driver with any unpaid charge cannot make a new reservation until the charge is paid at the counter.
BR-05	Cancelling 60 minutes or more before the start is free; later cancellation costs Rs 30; a no-show costs Rs 50.
BR-06	A reservation is held for 15 minutes after its start time, after which it expires.
BR-07	Charges are billed in 30-minute blocks with a minimum of 1 hour; overstay after a 10-minute grace period costs 1.5 times the hourly rate.
BR-08	A tariff change applies only to vehicles that enter after the change takes effect.
BR-09	A Lot Administrator may waive a charge only with a written reason, which is recorded in the audit log.

6.6 Domain Requirements

ID	Requirement
DR-01	Vehicle registration numbers follow the Indian formats defined by the Central Motor Vehicles Rules, 1989, including the Bharat (BH) series.
DR-02	All amounts are in Indian rupees with two decimal places, and all times are stored in UTC and shown in IST (UTC+05:30).
DR-03	Personal data is handled according to the Digital Personal Data Protection Act, 2023: it is collected only for parking services, and a Driver's personal data is deleted or anonymised within 30 days of account closure, except billing records kept for accounting.

7. Other Requirements

ID	Area	Requirement
OR-01	Database	PostgreSQL 16 with the btree_gist extension shall be the only system of record. Schema changes shall be made through versioned Django migrations, and the database shall be backed up daily with 7 days of backups kept.
OR-02	Localisation	The user interface shall be in English. Dates shall be shown as DD-MM-YYYY and times in 24-hour IST.
OR-03	Legal	Receipts shall state that they are not tax invoices, as GST invoicing is outside the scope of this release.
OR-04	Reuse	The system shall use Django's built-in authentication, session, CSRF and admin components; the team shall not write its own versions.

Appendix A: Glossary

Term	Definition
Allocation	The system's choice of a specific slot for a reservation or walk-in.
Booking reference	The unique code given to a reservation, in the form VPS-YYMMDD-NNNN.
CI	Continuous Integration: the automated build and test run on every push (GitHub Actions).
CSRF	Cross-Site Request Forgery: an attack that makes a signed-in user's browser send an unwanted request.
Distance rank	A number set per slot; 1 is the slot nearest to the exit.
DRF	Django REST Framework, used for the JSON endpoints.
Expiry job	The scheduled task that expires reservations not checked in within the hold window.
Hold window	The 15 minutes after a reservation's start time during which the vehicle can still check in.
HTMX	A small JavaScript library that lets server-rendered pages update parts of a page without a full reload.
IST	Indian Standard Time, UTC+05:30.
Lot	One car park, made up of slots.
No-show	A reservation that expires because the vehicle never checked in.
Overstay	Time parked after the reserved end time plus the 10-minute grace period.
RBAC	Role-based access control.
Slot	One parking bay, identified by a code within its lot.
SRS	Software Requirements Specification (this document).
STP	Software Test Plan.
TLS	Transport Layer Security, which encrypts HTTPS traffic.
UPI	Unified Payments Interface, an Indian instant payment system.
Walk-in	A vehicle that arrives without a reservation and is registered at the gate.

Appendix B: Field Layouts

B.1 Data Entry Fields

Table 1: Driver registration

Field	Length	Data Type	Description	Is Mandatory
Full name	100	String	Driver's name as entered at registration	Y
E-mail address	254	String	Sign-in identifier; unique	Y
Mobile number	10	Numeric	Indian mobile number starting with 6 to 9	Y
Password	8 to 128	String	Stored only as a hash (SEC-01)	Y
Accessible permit	1	Boolean	Set by a Lot Administrator (FR-AUTH-06)	N

Table 2: Vehicle

Field	Length	Data Type	Description	Is Mandatory
Registration number	13	Alphanumeric	Upper case, no spaces (FR-VEH-02)	Y
Vehicle type	12	Enumeration	Car, Two-wheeler or Electric car	Y
Nickname	30	String	Optional label shown to the Driver	N

Table 3: Reservation

Field	Length	Data Type	Description	Is Mandatory
Booking reference	15	Alphanumeric	VPS-YYMMDD-NNNN, unique	Y
Lot	-	Reference	Lot of the allocated slot	Y
Slot code	10	Alphanumeric	Allocated by the system (FR-ALLOC-02)	Y
Registration number	13	Alphanumeric	Vehicle for this reservation	Y
Start time	-	Date-time	Window start, IST on screen, UTC in storage	Y
End time	-	Date-time	Window end	Y
State	10	Enumeration	Reserved, Active, Completed, Cancelled or Expired	Y

Field	Length	Data Type	Description	Is Mandatory
Entry time	-	Date-time	Recorded at check-in	N
Exit time	-	Date-time	Recorded at check-out	N
Walk-in	1	Boolean	True when registered at the gate	Y

B.2 Report Requirements

Table 4: Fields included in each report

Occupancy Report (FR-RPT-02)	Revenue Report (FR-RPT-03)
Lot name	Lot name
Date	Date
Hour (00 to 23)	Slot type
Slot type	Number of completed stays
Total slots	Parking charges (Rs)
Average occupied slots	Overstay charges (Rs)
Average occupancy %	Late-cancellation and no-show charges (Rs)
Blocked slots	Payments by method (Cash, UPI, Card)
	Unpaid total (Rs)

B.3 Role Permission Matrix

Table 5: Functions allowed per role (enforced by SEC-03)

Function	Driver	Gate Attendant	Lot Administrator
Register own account, manage own vehicles	Yes	No	No
Search availability, view forecast	Yes	Yes	Yes
Create and cancel own reservations	Yes	No	No
Check in, check out, register walk-in, record payment	No	Yes	No
Correct entry or exit time	No	Yes	Yes
Manage lots, slots, blocks and tariffs	No	No	Yes
Manage staff accounts, set accessible permit	No	No	Yes
View reports and audit log, waive charges	No	No	Yes

Appendix C: Requirement Traceability Matrix

Architecture and design references point to the Software Architecture and Design Specification (SAD). Code file references are the planned module locations and will be updated as implementation proceeds. Test case IDs refer to the Software Test Plan; the last column lists the system and acceptance level tests among them.

Sl. No	Requirement ID	Brief Description of Requirement	Architecture Reference	Design Reference	Code File Reference	Test Case ID	System Test Case ID
1	FR-AUTH-01	Driver self-registration	Accounts and Access, Presentation layer	SAD 4.2 (Figure 4), 4.3	vps/accounts/, vps/*/views.py, templates/	TC-01, TC-02	TC-01
2	FR-AUTH-02	Sign in and sign out	Accounts and Access, Presentation layer	SAD 4.2 (Figure 4), 4.3	vps/accounts/, vps/*/views.py, templates/	TC-01, TC-03	TC-01
3	FR-AUTH-03	Three roles; staff created by admin	Accounts and Access, Presentation layer	SAD 4.2 (Figure 4), 4.3	vps/accounts/, vps/*/views.py, templates/	TC-21	-
4	FR-AUTH-04	Password reset link	Accounts and Access, Presentation layer	SAD 4.2 (Figure 4), 4.3	vps/accounts/, vps/*/views.py, templates/	TC-22	-
5	FR-AUTH-05	Deactivate staff accounts	Accounts and Access, Presentation layer	SAD 4.2 (Figure 4), 4.3	vps/accounts/, vps/*/views.py, templates/	TC-21	-
6	FR-AUTH-06	Accessible-parking permit flag	Accounts and Access, Presentation layer	SAD 4.2 (Figure 4), 4.3	vps/accounts/, vps/*/views.py, templates/	TC-23	-
7	FR-VEH-01	Up to 5 vehicles per Driver	Vehicles	SAD 4.3, 4.6	vps/vehicles/	TC-24	-
8	FR-VEH-02	Registration number format	Vehicles	SAD 4.3, 4.6	vps/vehicles/	TC-04	-
9	FR-VEH-03	Unique registration number	Vehicles	SAD 4.3, 4.6	vps/vehicles/	TC-04	-
10	FR-VEH-04	Protect vehicles with bookings	Vehicles	SAD 4.3, 4.6	vps/vehicles/	TC-24	-
11	FR-LOT-01	Manage lots	Lots and Slots, Notifications	SAD 4.3, 4.6	vps/lots/, vps/notifications/	TC-25	-
12	FR-LOT-02	Manage slots	Lots and Slots, Notifications	SAD 4.3, 4.6	vps/lots/, vps/notifications/	TC-25	-
13	FR-LOT-03	Block slot for maintenance	Lots and Slots, Notifications	SAD 4.3, 4.6	vps/lots/, vps/notifications/	TC-05	-
14	FR-LOT-04	Move bookings off blocked slot	Lots and Slots, Notifications	SAD 4.3, 4.6	vps/lots/, vps/notifications/	TC-26	-
15	FR-SRCH-01	Availability search	Slot Allocation, Presentation layer	SAD 4.3	vps/allocation/, vps/*/views.py, templates/	TC-05	-
16	FR-SRCH-02	Search window validation	Slot Allocation, Presentation layer	SAD 4.3	vps/allocation/, vps/*/views.py, templates/	TC-27	-
17	FR-SRCH-03	Live occupancy view	Slot Allocation, Presentation layer	SAD 4.3	vps/allocation/, vps/*/views.py, templates/	TC-28	TC-28
18	FR-RES-01	Create reservation	Reservations, Slot Allocation, Scheduler, Notifications, PostgreSQL database	SAD 4.2 (Figure 2), 4.3, 4.6	vps/reservations/, vps/allocation/, vps/reservations/jobs.py, vps/notifications/	TC-06	-

Sl. No	Requirement ID	Brief Description of Requirement	Architecture Reference	Design Reference	Code File Reference	Test Case ID	System Test Case ID
					vps/*/migrations/		
19	FR-RES-02	No double booking	Reservations, Slot Allocation, Scheduler, Notifications, PostgreSQL database	SAD 4.2 (Figure 2), 4.3, 4.6	vps/reservations/, vps/allocation/, vps/reservations/jobs.py, vps/notifications/, vps/*/migrations/	TC-07	-
20	FR-RES-03	Reservation limit and unpaid charges	Reservations, Slot Allocation, Scheduler, Notifications, PostgreSQL database	SAD 4.2 (Figure 2), 4.3, 4.6	vps/reservations/, vps/allocation/, vps/reservations/jobs.py, vps/notifications/, vps/*/migrations/	TC-08	-
21	FR-RES-04	Cancellation and late fee	Reservations, Slot Allocation, Scheduler, Notifications, PostgreSQL database	SAD 4.2 (Figure 2), 4.3, 4.6	vps/reservations/, vps/allocation/, vps/reservations/jobs.py, vps/notifications/, vps/*/migrations/	TC-09	-
22	FR-RES-05	Expiry and no-show fee	Reservations, Slot Allocation, Scheduler, Notifications, PostgreSQL database	SAD 4.2 (Figure 2), 4.3, 4.6	vps/reservations/, vps/allocation/, vps/reservations/jobs.py, vps/notifications/, vps/*/migrations/	TC-10	-
23	FR-RES-06	Booking history	Reservations, Slot Allocation, Scheduler, Notifications, PostgreSQL database	SAD 4.2 (Figure 2), 4.3, 4.6	vps/reservations/, vps/allocation/, vps/reservations/jobs.py, vps/notifications/, vps/*/migrations/	TC-29	-
24	FR-RES-07	Booking e-mails	Reservations, Slot Allocation, Scheduler, Notifications, PostgreSQL database	SAD 4.2 (Figure 2), 4.3, 4.6	vps/reservations/, vps/allocation/, vps/reservations/jobs.py, vps/notifications/, vps/*/migrations/	TC-10, TC-26	-
25	FR-ALLOC-01	Slot eligibility rules	Slot Allocation, PostgreSQL database	SAD 4.2 (Figure 2), 4.6	vps/allocation/, vps/*/migrations/	TC-06, TC-23	-
26	FR-ALLOC-02	Deterministic slot choice	Slot Allocation, PostgreSQL database	SAD 4.2 (Figure 2), 4.6	vps/allocation/, vps/*/migrations/	TC-06	-
27	FR-ALLOC-03	Retry on conflict	Slot Allocation, PostgreSQL database	SAD 4.2 (Figure 2), 4.6	vps/allocation/, vps/*/migrations/	TC-07	-
28	FR-GATE-01	Check-in window	Gate Operations, Billing, Audit Log	SAD 4.2 (Figure 3), 4.3	vps/gate/, vps/billing/, vps/audit/	TC-11	TC-11

Sl. No	Requirement ID	Brief Description of Requirement	Architecture Reference	Design Reference	Code File Reference	Test Case ID	System Test Case ID
29	FR-GATE-02	Early arrival refused	Gate Operations, Billing, Audit Log	SAD 4.2 (Figure 3), 4.3	vps/gate/, vps/billing/, vps/audit/	TC-11	TC-11
30	FR-GATE-03	Walk-in registration	Gate Operations, Billing, Audit Log	SAD 4.2 (Figure 3), 4.3	vps/gate/, vps/billing/, vps/audit/	TC-12	TC-12
31	FR-GATE-04	Check-out and receipt	Gate Operations, Billing, Audit Log	SAD 4.2 (Figure 3), 4.3	vps/gate/, vps/billing/, vps/audit/	TC-13	-
32	FR-GATE-05	Record payment	Gate Operations, Billing, Audit Log	SAD 4.2 (Figure 3), 4.3	vps/gate/, vps/billing/, vps/audit/	TC-30	-
33	FR-GATE-06	Manual time correction	Gate Operations, Billing, Audit Log	SAD 4.2 (Figure 3), 4.3	vps/gate/, vps/billing/, vps/audit/	TC-31	-
34	FR-GATE-07	Reallocate occupied slot at check-in	Gate Operations, Billing, Audit Log	SAD 4.2 (Figure 3), 4.3	vps/gate/, vps/billing/, vps/audit/	TC-32	-
35	FR-BILL-01	Tariff configuration	Billing	SAD 4.2 (Figure 3), 4.6	vps/billing/	TC-33	-
36	FR-BILL-02	Parking charge calculation	Billing	SAD 4.2 (Figure 3), 4.6	vps/billing/	TC-13	-
37	FR-BILL-03	Overstay charge	Billing	SAD 4.2 (Figure 3), 4.6	vps/billing/	TC-14	-
38	FR-BILL-04	Receipt contents	Billing	SAD 4.2 (Figure 3), 4.6	vps/billing/	TC-13	-
39	FR-BILL-05	Unpaid charges shown	Billing	SAD 4.2 (Figure 3), 4.6	vps/billing/	TC-30	-
40	FR-RPT-01	Live occupancy dashboard	Reports, Audit Log	SAD 4.3	vps/reports/, vps/audit/	TC-28	TC-28
41	FR-RPT-02	Occupancy report	Reports, Audit Log	SAD 4.3	vps/reports/, vps/audit/	TC-34	-
42	FR-RPT-03	Revenue report	Reports, Audit Log	SAD 4.3	vps/reports/, vps/audit/	TC-34	-
43	FR-RPT-04	Audit log search	Reports, Audit Log	SAD 4.3	vps/reports/, vps/audit/	TC-19	-
44	FR-FCST-01	Availability forecast	Forecast service, Model trainer, Scheduler	SAD 4.3	forecast/app/, forecast/train/, vps/reservations/jobs.py	TC-35	-
45	FR-FCST-02	Weekly retraining and baseline	Forecast service, Model trainer, Scheduler	SAD 4.3	forecast/app/, forecast/train/, vps/reservations/jobs.py	TC-35	-
46	FR-FCST-03	Forecast fallback	Forecast service, Model trainer, Scheduler	SAD 4.3	forecast/app/, forecast/train/, vps/reservations/jobs.py	TC-20	-
47	NFR-PERF-01	Search response time	Presentation layer, PostgreSQL database	SAD 4.4	vps/*/views.py, templates/, vps/*/migrations/	TC-18	TC-18
48	NFR-PERF-02	Booking and gate response time	Presentation layer, PostgreSQL database	SAD 4.4	vps/*/views.py, templates/, vps/*/migrations/	TC-36	TC-36
49	NFR-PERF-03	200 concurrent sessions	Presentation layer, PostgreSQL	SAD 4.4	vps/*/views.py, templates/,	TC-18	TC-18

Sl. No	Requirement ID	Brief Description of Requirement	Architecture Reference	Design Reference	Code File Reference	Test Case ID	System Test Case ID
			database		vps/*/migrations/		
50	NFR-PERF-04	Page load time and size	Presentation layer, PostgreSQL database	SAD 4.4	vps/*/views.py, templates/, vps/*/migrations/	TC-37	TC-37
51	NFR-PERF-05	Forecast response time	Forecast service	SAD 4.4	forecast/app/	TC-36	TC-36
52	SAF-01	Blocked slots never allocated	Lots and Slots, Slot Allocation	SAD 3.9, 4.4	vps/lots/, vps/allocation/	TC-05	-
53	SAF-02	Accessible slots protected	Slot Allocation	SAD 3.9, 4.4	vps/allocation/	TC-23	-
54	SAF-03	Manual fallback at the gate	Gate Operations	SAD 3.9, 4.4	vps/gate/	TC-31	-
55	SEC-01	Password hashing and policy	Accounts and Access	SAD 3.9, 4.4	vps/accounts/	TC-39	-
56	SEC-02	Lockout and rate limit	Accounts and Access	SAD 3.9, 4.4	vps/accounts/	TC-03	-
57	SEC-03	Server-side role checks	Presentation layer	SAD 3.9, 4.4	vps/*/views.py, templates/	TC-15	-
58	SEC-04	HTTPS and secure cookies	Reverse proxy	SAD 3.9, 4.4	deploy/nginx/	TC-38	TC-38
59	SEC-05	CSRF protection	Presentation layer	SAD 3.9, 4.4	vps/*/views.py, templates/	TC-16	-
60	SEC-06	Input validation, parameterised queries	Presentation layer, PostgreSQL database	SAD 3.9, 4.4	vps/*/views.py, templates/, vps/*/migrations/	TC-17	TC-17
61	SEC-07	Audit log	Audit Log	SAD 3.9, 4.4	vps/audit/	TC-19, TC-31	-
62	SEC-08	Idle timeout, session rotation	Accounts and Access	SAD 3.9, 4.4	vps/accounts/	TC-38	TC-38
63	SEC-09	Masking; no secrets on screen or in logs	Presentation layer	SAD 3.9, 4.4	vps/*/views.py, templates/	TC-22, TC-39	-
64	SEC-10	Secrets outside the repository	CI pipeline, All components	SAD 3.9, 4.4	.github/workflows/ci.yml, (whole repository)	TC-40	TC-40
65	NFR-REL-01	Exactly one winner for last slot	Slot Allocation, PostgreSQL database	SAD 4.4	vps/allocation/, vps/*/migrations/	TC-07	-
66	NFR-AVL-01	99% availability in operating hours	Reverse proxy, All components	SAD 4.4	deploy/nginx/, (whole repository)	Uptime monitoring (STP 14)	Uptime monitoring
67	NFR-USE-01	First booking under 3 minutes	Presentation layer	SAD 4.4	vps/*/views.py, templates/	TC-41	TC-41
68	NFR-USE-02	Small screens and contrast	Presentation layer	SAD 4.4	vps/*/views.py, templates/	TC-42	TC-42
69	NFR-MNT-01	Test coverage thresholds	CI pipeline	SAD 4.4	.github/workflows/ci.yml	TC-43	TC-43
70	NFR-MNT-02	Lint and reviewed merges	CI pipeline	SAD 4.4	.github/workflows/ci.yml	TC-43	TC-43
71	NFR-PRT-01	Server and browser support	All components	SAD 4.4	(whole repository)	TC-42	TC-42
72	NFR-TST-01	Rules testable without web layer	Slot Allocation, Billing	SAD 4.4	vps/allocation/, vps/billing/	TC-43	TC-43